

Bit Packing — Super Detailed Guide (Kid Friendly)

1. What is the Goal?

We want to store multiple numbers inside ONE number. We do this using binary (0 and 1). Think of it like packing things into a box.

2. Binary Basics

Every number in a computer is stored in binary. Example: 1 = 1 2 = 10 3 = 11 4 = 100 5 = 101 15 = 1111

3. VERY IMPORTANT Example: 15 << 7

We will break this step by step below.

```
Step 1: Convert to binary
15 = 1111

Step 2: Shift left by 7
Add 7 zeros:

1111 << 7 = 1111 0000000

Step 3: Final binary
11110000000

Step 4: Convert to decimal
= 1×210 + 1×29 + 1×28 + 1×27
= 1024 + 512 + 256 + 128
= 1920

FINAL:
15 << 7 = 1920
```

4. Right Shift Example: 15 >> 2

Now we move bits to the right.

```
Step 1:
15 = 1111

Step 2:
1111 >> 2 = remove last 2 bits = 11

Step 3:
11 = 3

FINAL:
15 >> 2 = 3
```

5. Packing Two Numbers

We combine two numbers into one.

```
region_id = 3
user_id = 25

Step 1:
3  = 00000011
25 = 00011001

Step 2:
3 << 8 = 00000011 00000000

Step 3: OR
00000011 00000000
|00000000 00011001
-----
00000011 00011001

Step 4:
= 793
```

6. Extract user_id

Use AND with mask.

```
combined = 00000011 00011001
mask      = 00000000 11111111

AND:
= 00000000 00011001 = 25
```

7. Extract region_id

Use right shift.

```
00000011 00011001 >> 8
= 00000011
= 3
```

8. Why $(1 \ll n) - 1$

This creates n number of 1s.

```
Example: n = 5

1 << 5 = 100000
minus 1:
100000
-      1
-----
011111

= 31
```

9. Character Example

Characters are numbers inside computers.

'A' = 65 = 01000001

Step:

65 << 8 = 01000001 00000000

OR 5:

= 01000001 00000101

Decode:

>> 8 → 65 → 'A'

10. String Example

Pack two characters.

'H' = 72 = 01001000

'i' = 105 = 01101001

Pack:

01001000 << 8 = 01001000 00000000

OR:

= 01001000 01101001

Decode:

last 8 bits → i

shift >> 8 → H

11. Golden Rule

<< = make space, | = fill, >> = bring back, & = extract exact bits